

oldver\config3.py

```
1
2 import os
3 import random
4 import numpy as np
5 import torch
6 from pathlib import Path
7
8 # — Reproducibility —————
9 SEED = 42
10
11 def seed_everything(seed: int = SEED):
12     """Lock every RNG source so a re-run reproduces identical results."""
13     random.seed(seed)
14     np.random.seed(seed)
15     torch.manual_seed(seed)
16     torch.cuda.manual_seed(seed)
17     torch.cuda.manual_seed_all(seed)
18     torch.backends.cudnn.deterministic = True
19     torch.backends.cudnn.benchmark = False
20     os.environ["PYTHONHASHSEED"] = str(seed)
21
22 # — Paths (Ex3!) —————
23 BASE_DIR = Path(r"C:\Users\sammy\Exam3Ex3")
24 DATA_DIR = BASE_DIR / "vir_data_exam"
25 TRAIN_DIR = DATA_DIR / "train"
26 TEST_DIR = DATA_DIR / "test1_vir"
27 OUTPUT_DIR = BASE_DIR / "outputs"
28 FIGURES_DIR = OUTPUT_DIR / "figures"
29 OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
30 FIGURES_DIR.mkdir(parents=True, exist_ok=True)
31
32 # — Classes —————
33 CLASS_NAMES = ["class 1", "class 2", "class 3"]
34 NUM_CLASSES = len(CLASS_NAMES)
35 CLASS_TO_IDX = {name: idx for idx, name in enumerate(CLASS_NAMES)}
36 IDX_TO_CLASS = {idx: name for idx, name in enumerate(CLASS_NAMES)}
37
38 # — Training hyperparameters —————
39 IMAGE_SIZE = 224
40 BATCH_SIZE = 32
41 GRAD_ACCUM_STEPS = 1
42 EPOCHS = 60
43 WARMUP_EPOCHS = 5
44 LEARNING_RATE = 3e-4
45 WEIGHT_DECAY = 1e-4
46 PATIENCE = 12
47 VAL_SPLIT = 0.2
48 LABEL_SMOOTHING = 0.1
49 DROPOUT = 0.4
50
51 # — Architecture feature flags —————
52 USE_SE = True
53 SE_REDUCTION = 16
54 USE_AUX = True
55 AUX_WEIGHT = 0.3
56 COMPARE_SE = True
57
58 # — Mixup —————
59 USE_MIXUP = False
60 MIXUP_ALPHA = 0.2
61
62 # — Augmentation strength —————
63 COLOR_JITTER = (0.3, 0.3, 0.3, 0.10)
64 CHANNEL_SCALE = 0.15
65 NOISE_STD = 0.04
66
67 # — DataLoader workers —————
68 NUM_WORKERS = 2
69
70 # — Device —————
```

```
71 | DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
72 |
73 | # — Output file paths —————
74 | MODEL_SAVE_PATH      = OUTPUT_DIR / "report3.pth"
75 | MODEL_SAVE_PATH_ROOT = BASE_DIR   / "report3.pth"
76 | HISTORY_SAVE_PATH    = OUTPUT_DIR / "training_history.csv"
77 | UTIL_SAVE_PATH       = BASE_DIR   / "util3.txt"
78 | STATS_PATH           = OUTPUT_DIR / "train_stats.json"
```